

7. Interdisciplinary Modeling in Quantitative Trading (Part 2/3)

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1. Co-movements in Finance

Co-movement, in Finance, refers to the tendency of two or more asset prices, returns, or financial features to move together over time, reflecting underlying statistical connections.

Key dimensions of modeling and trading co-movements:
causal logic, directionality, strength, and persistence.

Key assumptions of quant strategies based on co-movements:

- **Convergence vs Divergence:** The identified mispricing or divergence is temporary and will revert to its historical mean or fair value.
- **Statistical Stability:** The historical statistical relationship (correlation, cointegration, beta, etc.) between the paired assets is assumed to persist into the future.
- **Execution & No Market Impact:** The model assumes the assets can be bought and sold in the required quantities thus preserving the profitability of the arbitrage.

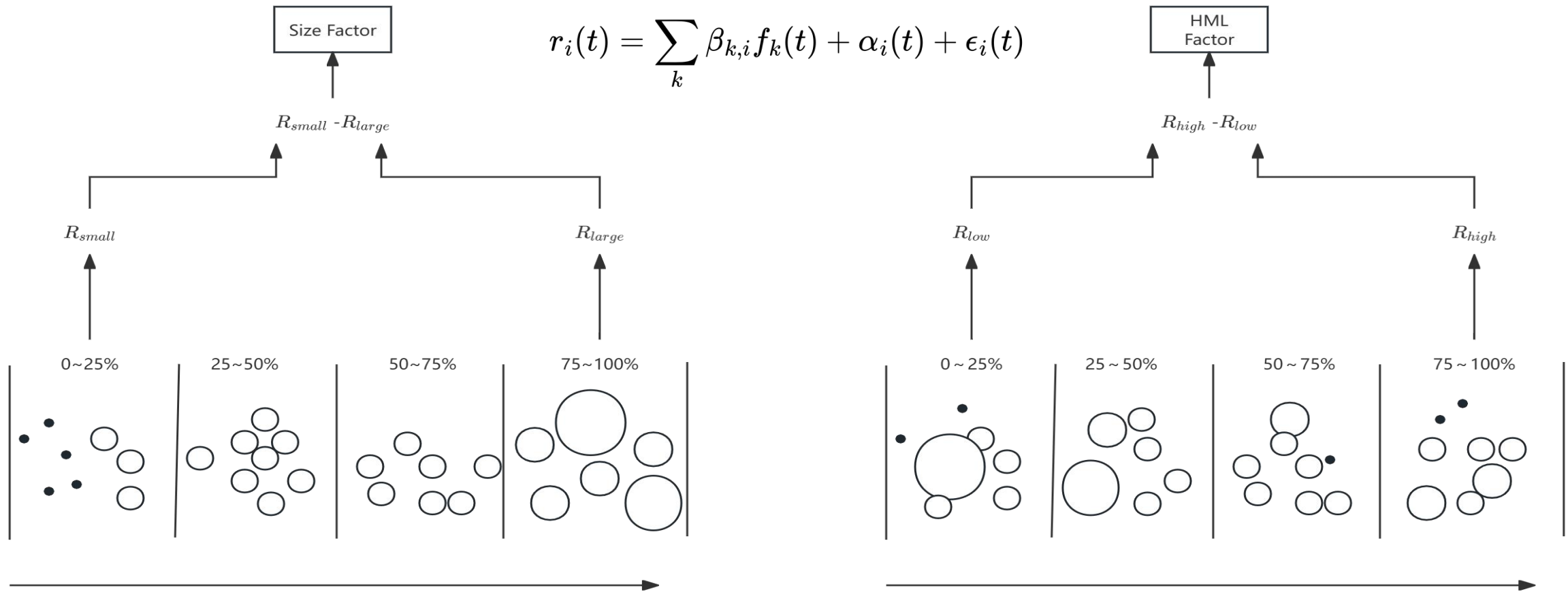


statistical_arbitrage
high-frequency_arbitrage
market_neutral
relative_value
alpha_strategies
pairs_trading
dollar-neutral
delta_neutral
beta_neutral
hedging

1. Co-movements in Finance: Types of Co-movements and Related Strategies

Scenario	Strategy/Pairs	Model Complexity	Frequency	Market Risk	Execution Risk
Same Underlying, Different Markets	Cross-Exchange/Market Arbitrage - Stock-ADR (American Depositary Receipts) - Crypto-Crypto - Stock-Tokenized Stock - Index ETF-Index CFD	Low	High	Low	High
	Cross-Asset Arbitrage - Stock-ETF (ETF Redemption Arb.) - FX-FX	Low	High	Low	High
Same Underlying, Different Assets	Underlying-Derivatives Relative Value - Stock-Option - Stock-Convertible Bonds - Commodity-Future - Index ETF-Index Options	High	Low-High	Medium-High	Low-Medium
	Relative Value (Not Arbitrage) - Stock-Stock - Stock-ETF - Commodity-Commodity - ETF-ETF	Medium	Low-High	Medium-High	Depends on Frequency
Different Underlyings, Same Market	Relative Value (Not Arbitrage) - Fundamental (BTC-MSTR) - Supply Chain (AAPL-002475.SZ) - Global-Macro	High	Low-Medium	High	Low
	Pairs Trading (Not Arbitrage) Statistical Pairs	High	Low-High	High	Depends on Frequency

2. Co-movements based Quantitative Strategies: Factor Model: What are the factors?



- The return of assets are driven by certain factors.
- Most popular factors are the average return premium between two groups of assets.
- Factors may have intersections of their component assets.

2. Co-movements based Quantitative Strategies: Factor Model, Beta Hedging

Identify Pairs:

Economic Logic
Different Assets
Same Market



Calculate Rolling Beta:

$$r_1(t - N, t) = \beta(t)r_2(t - N, t) + \alpha(t)$$

Optimize:

$$\bar{\beta} = \frac{1}{n} \sum \beta(t - n, t)$$

$$\alpha(t) = r_1(t - N, t) - \bar{\beta}r_2(t - N, t)$$

Filter:

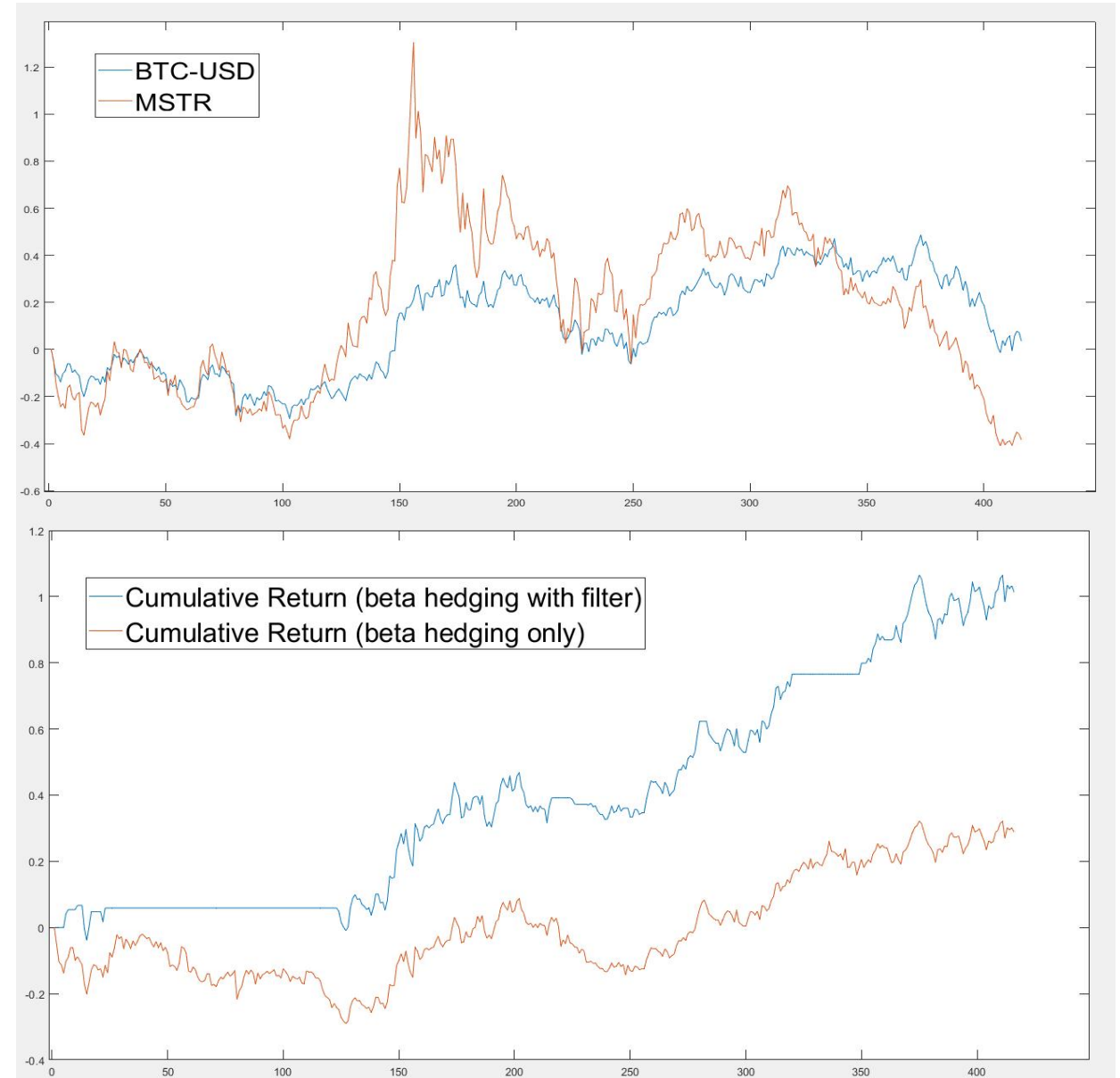
If $\alpha(t) < k\%$, close positions.

α has its momentum to stay positive/negative.



Trading Rules/Execution:

Buy 1 share IBIT, short $\bar{\beta}$ shares of MSTR.
Rebalance every day at close.



2. Co-movements based Quantitative Strategies: Factor Model, Multi-Factor Model

- In Finance, factor models serve as the primary tool for identifying and exploiting predictable patterns in market prices.
- **Factor models are designed more for explanatory power than predictive power, serving to decompose past returns rather than reliably foresee future ones.**

$$r_i(t) = \sum_k \beta_{k,i} f_k(t) + \alpha_i(t) + \epsilon_i(t)$$

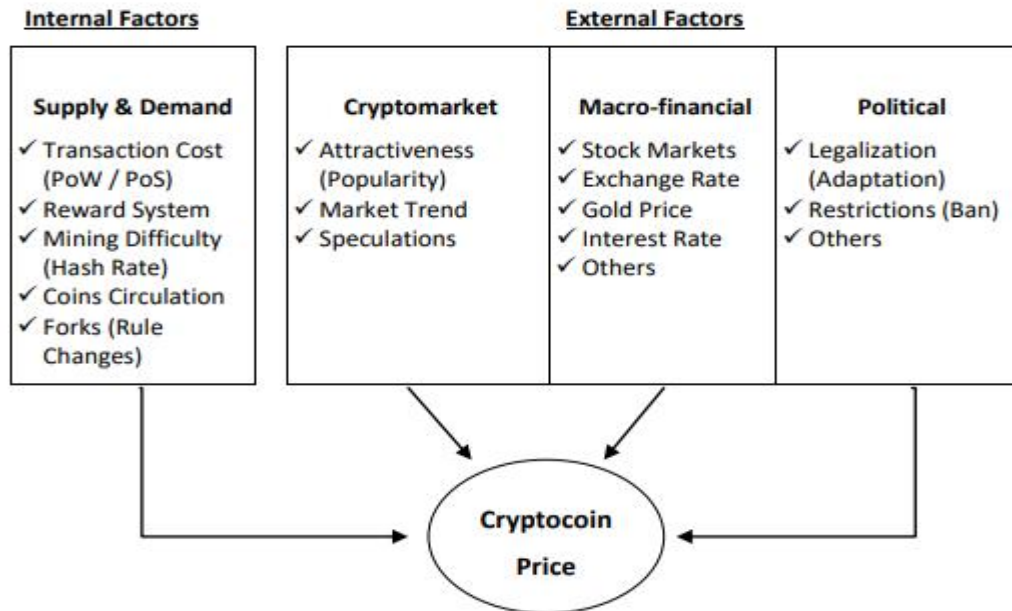
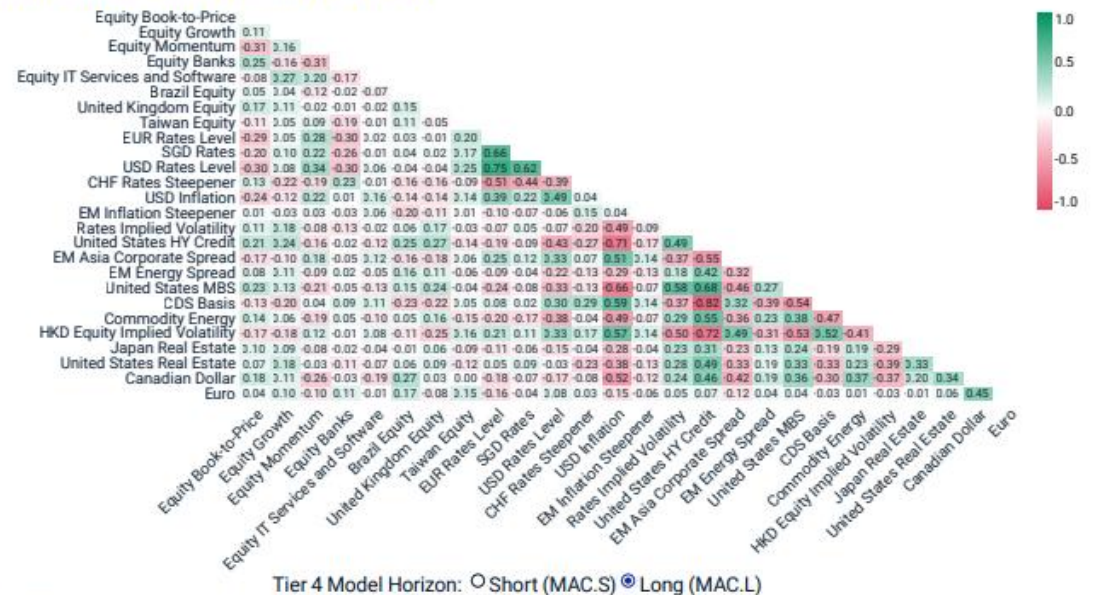


Figure 1. Factors that Influence Cryptocurrency Prices

Tier 4 Factor Correlations (as of May 29, 2020)



Source: The MSCI Multi-Asset Class Factor Model (MAC).

- Machine learning factor miner: Feng, G., Giglio, S., & Xiu, D. (2020). Taming the factor zoo: A test of new factors. The Journal of Finance, 75(3), 1327-1370.
- Well defined factor lists (400+): Hou, K., Xue, C., & Zhang, L. (2017). A comparison of new factor models. Fisher college of business working paper, (2015-03), 05.
- Factors of Cryptocurrency: Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. The Journal of Finance, 77(2), 1133-1177.
- Factors of Cryptocurrency: Sovbetov, Y. (2025). Factors influencing cryptocurrency prices: Evidence from bitcoin, ethereum, dash, litecoin, and monero. arXiv preprint arXiv:2511.22782.

2. Co-movements based Quantitative Strategies: Factor Model, Market Neutral

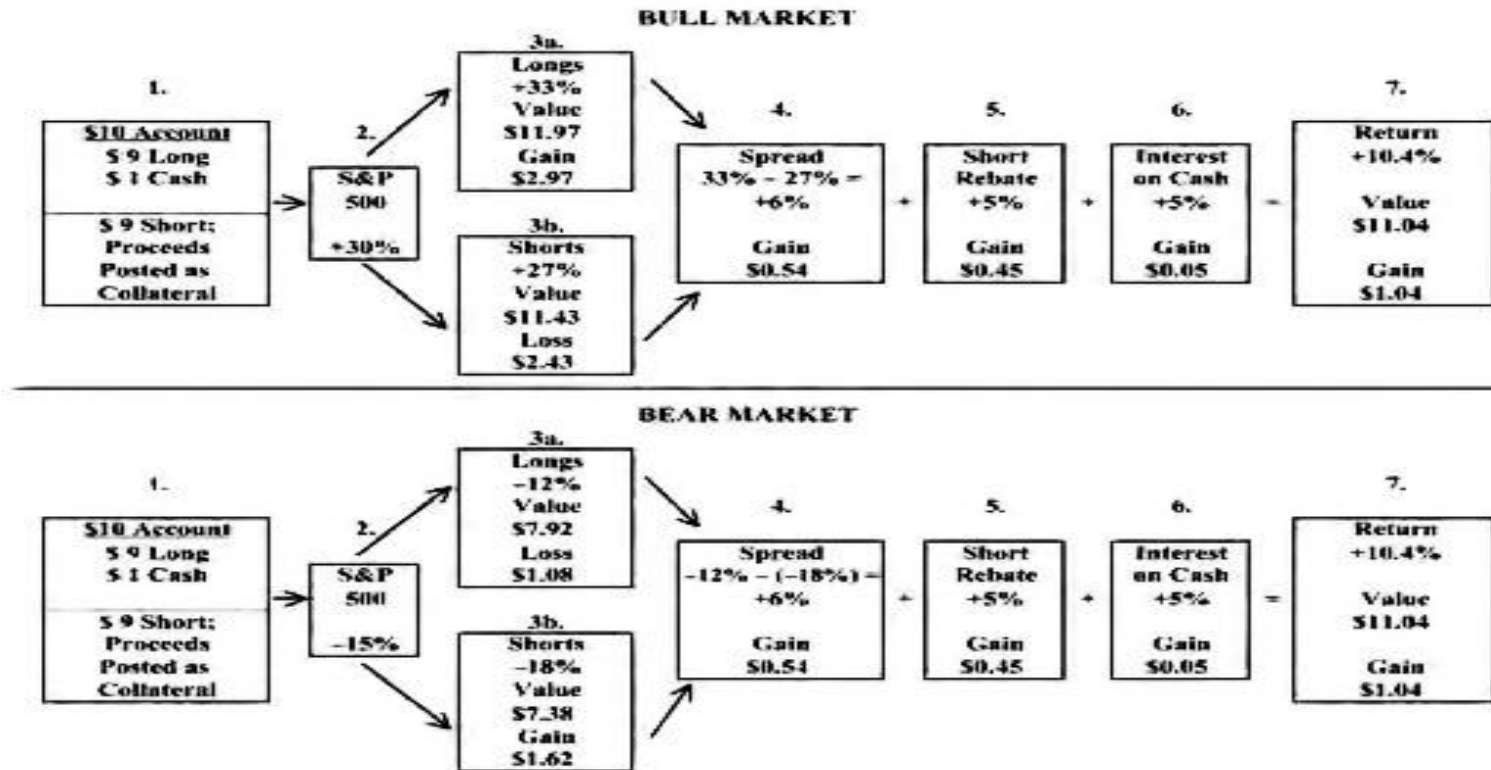
- Factor models help to assess the difference between assets in the scope of selected factors.

$$R_1 = 1.2Factor_{market} + 1.7Factor_{Size} + 1.5Factor_{HML} + \alpha_1 + \epsilon_1$$

$$R_2 = 0.8Factor_{market} + 0.6Factor_{Size} + 0.25Factor_{HML} + \alpha_2 + \epsilon_2$$

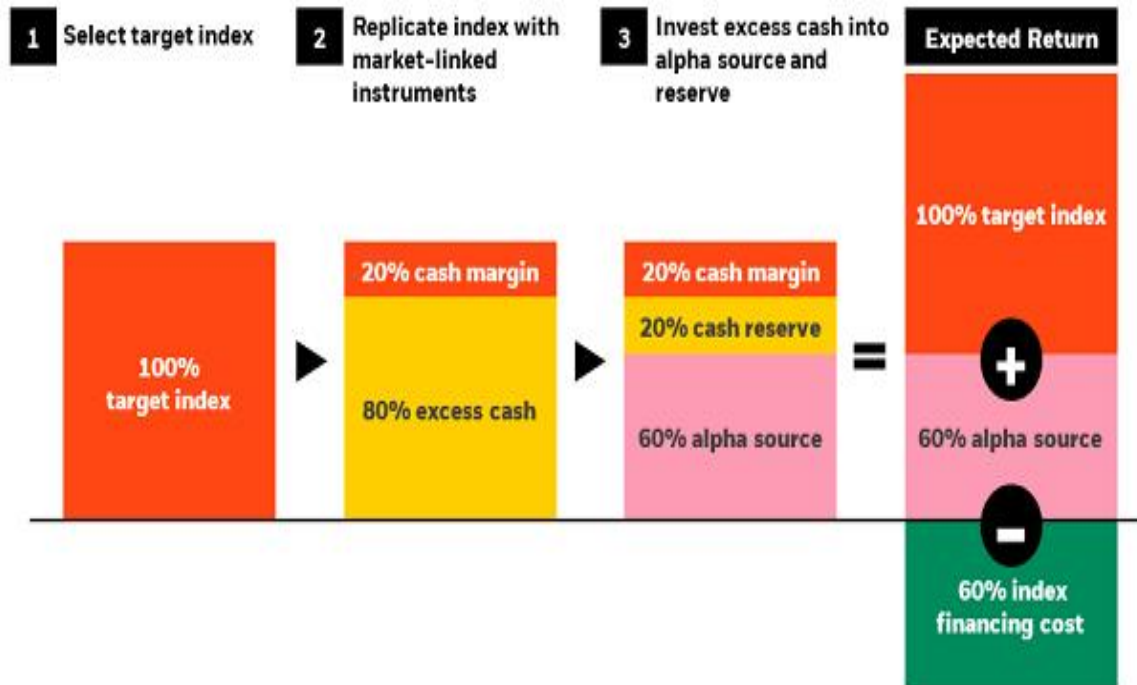
- Hedged Portfolio, 0 exposure to the market.

$$R_1 - 1.5R_2 = 0Factor_{Market} + 0.8Factor_{Size} + 1.125Factor_{HML} + \alpha_1 - 1.5\alpha_2$$



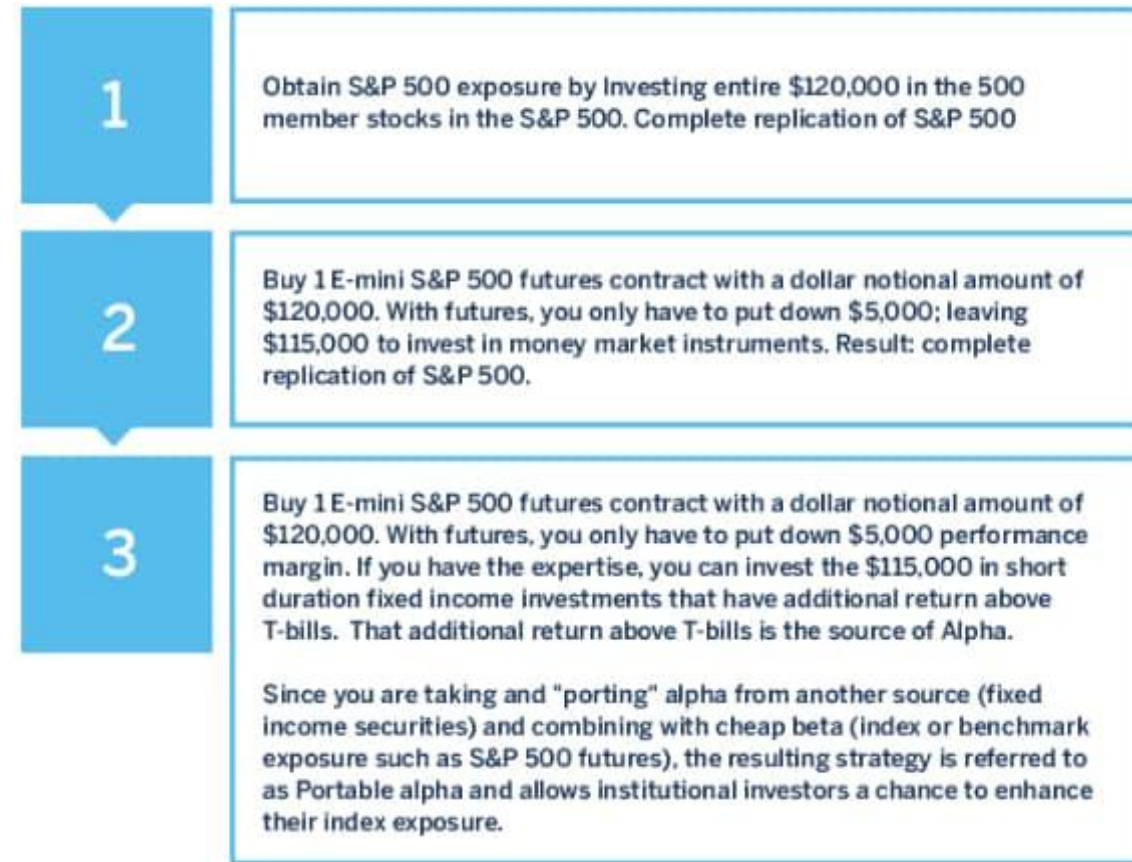
2. Co-movements based Quantitative Strategies: Factor Model, Market Neutral, Portable α

Financial Engineering for the Portable α



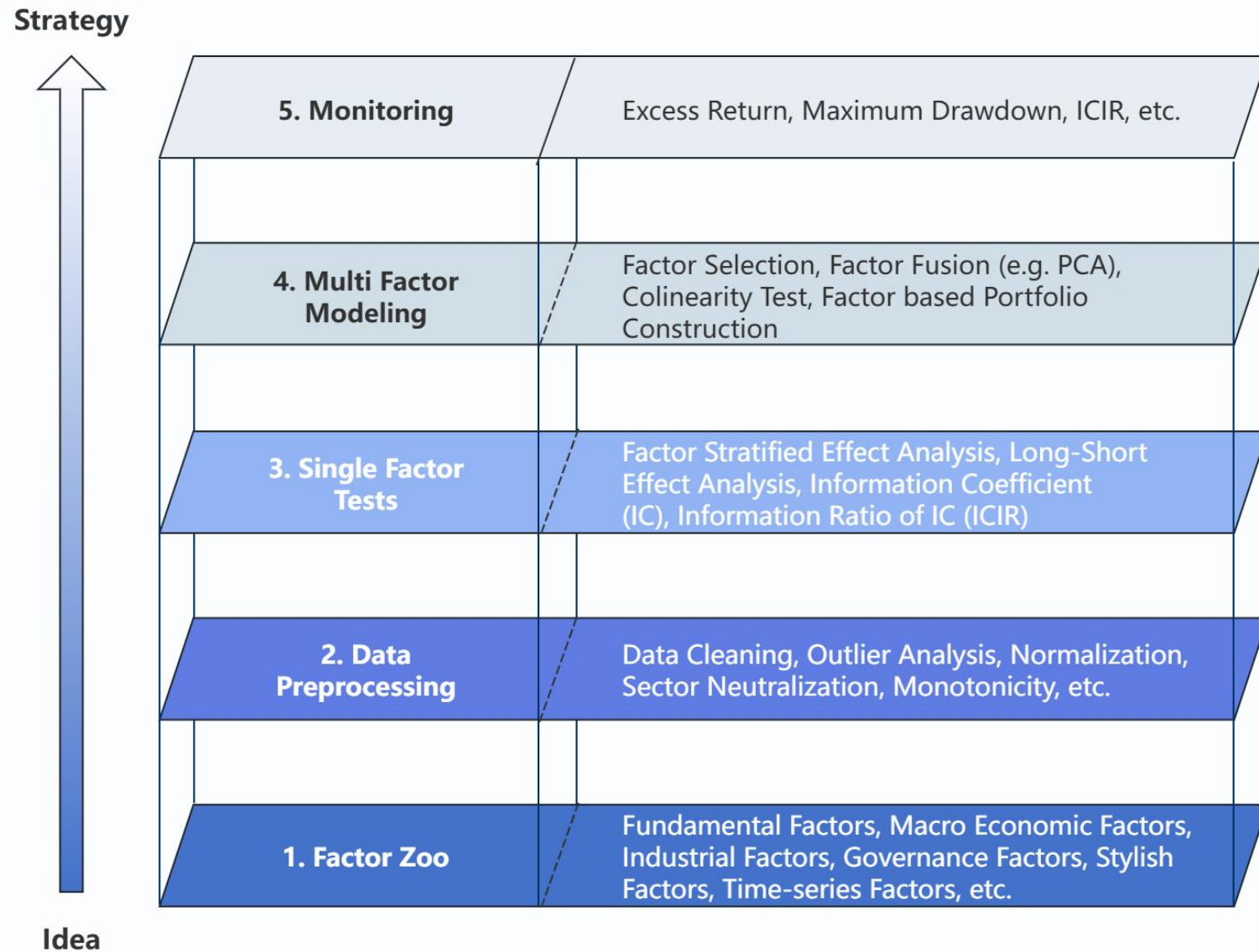
source: Black Rock

Portable α : Example with E-mini Future

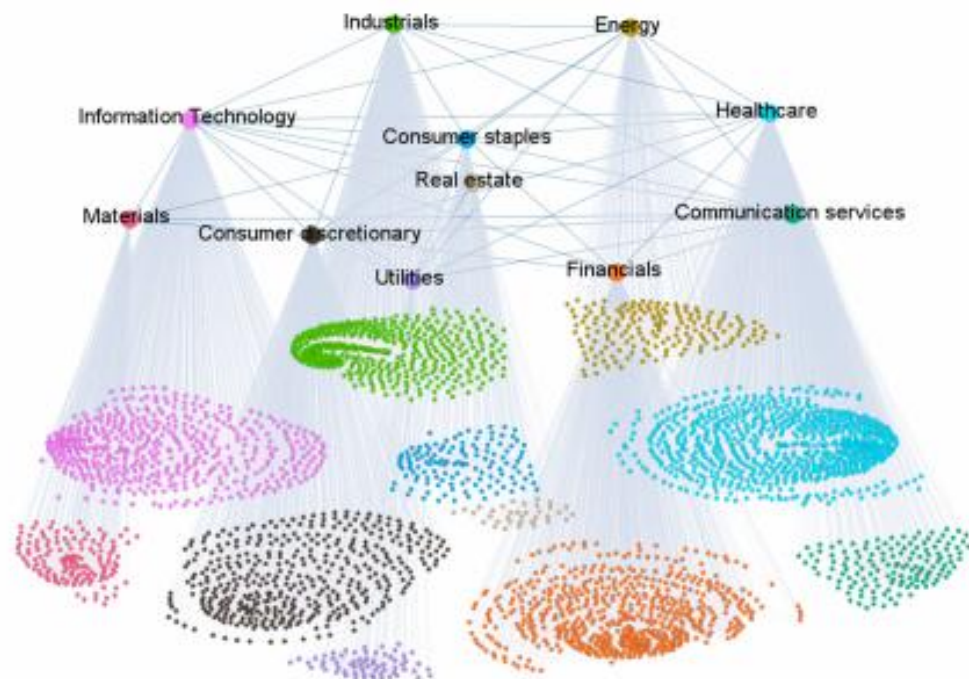


source: CME

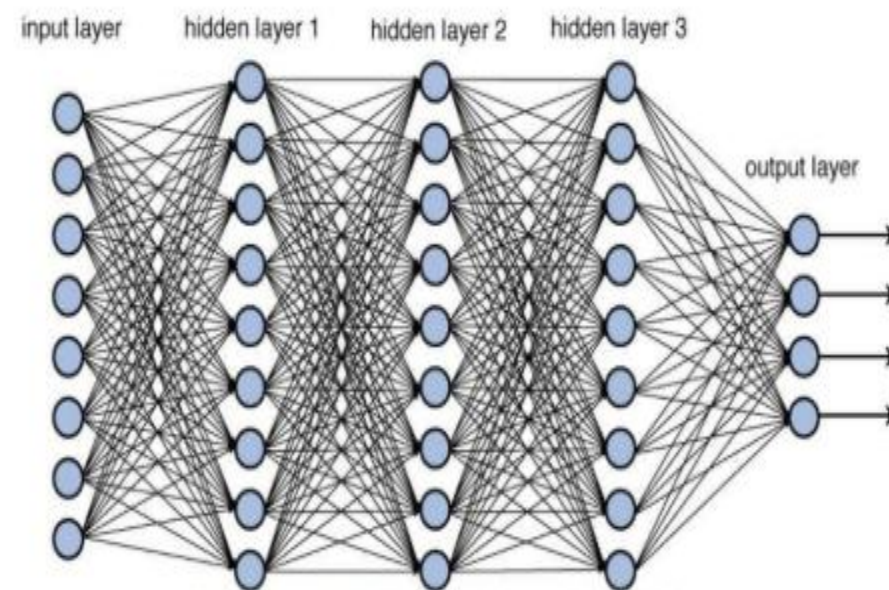
2. Co-movements based Quantitative Strategies: Factor Model, A Typical Workflow



2. Co-movements based Quantitative Strategies: Factor Model, Alternative Ways for Factor Modeling



Complex Network



Machine Learning

2. Co-movements based Quantitative Strategies: Factor Model, Shapley Values

Shapley Value formula

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)]$$

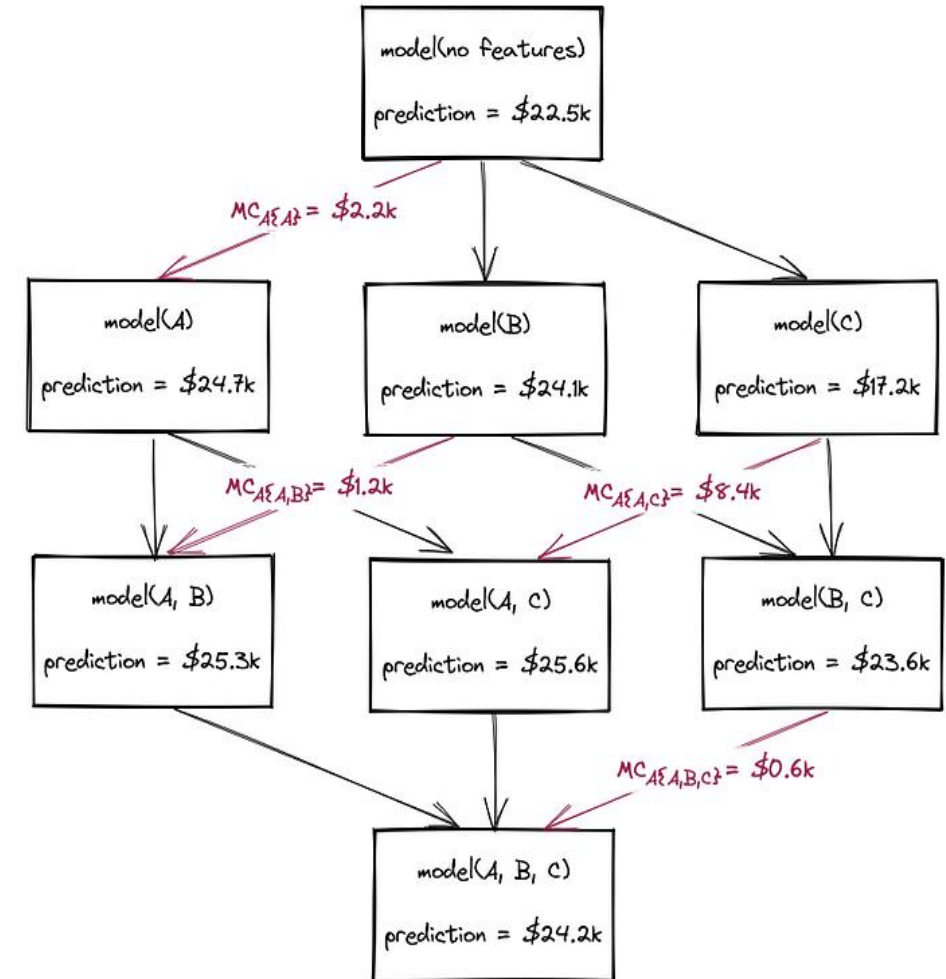
Weighted average: weight comes from the probability that the player make some contribution

Prediction with the feature included

Prediction with the feature removed

Sum over all possible coalition player i can join

Marginal contribution of a feature i to coalition S

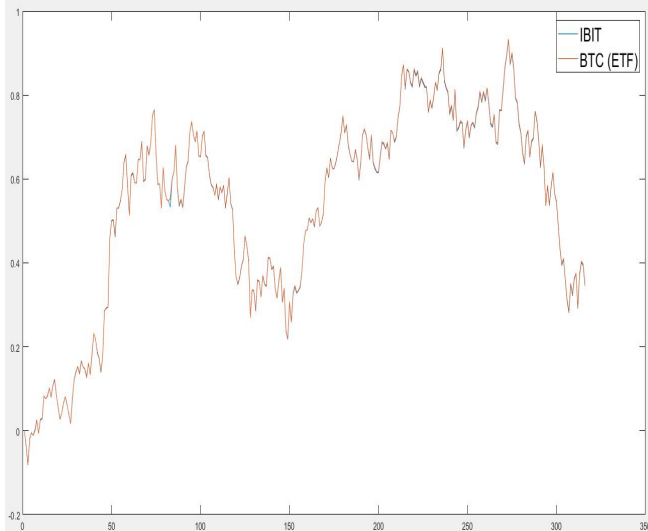


2. Co-movements based Quantitative Strategies: Cross-Asset Arbitrage, High-Frequency Arbitrage

IBIT-BTC (ETF Symbol) spread >0 , short IBIT, long BTC with equal value; close when spread ≤ 0 ;
IBIT-BTC (ETF Symbol) spread <0 , long IBIT, short BTC with equal value; close when spread ≤ 0 ;

Identify Pairs:

Same Underlying
Same Asset Class
Same Market
Similar Expense Ratio

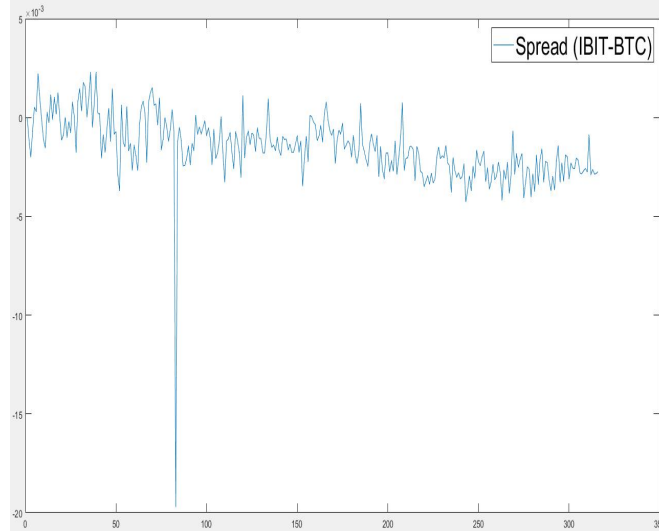


Calculate Spread:

Closing Price Spread
Bid-Ask Spread

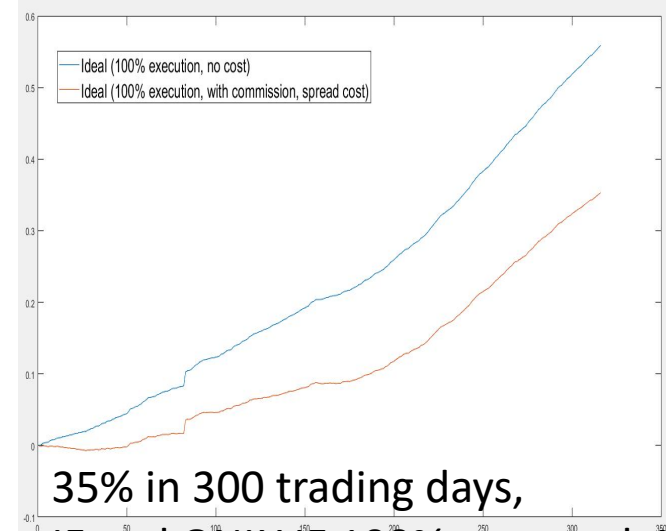
Optimize Order:

Price/Type(Fill or Kill)/Size, etc.



Execution:

Send orders to the exchange,
Check execution,
Check closing signals,
Send close orders
Check execution, ...
Manage PnL, ...



35% in 300 trading days,
IF and ONLY IF 100% executed, NO slippage!

2. Co-movements based Quantitative Strategies: Cross-Asset Arbitrage, Triangular Arbitrage

Triangular Arbitrage:

Identify Pairs:

Same Underlying
Same Asset Class
Same Market



Calculate Spread:

Bid-Ask Spread

Optimize Order:

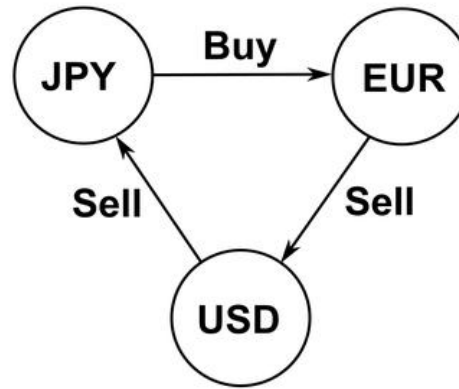
Price/Type/Size, etc.



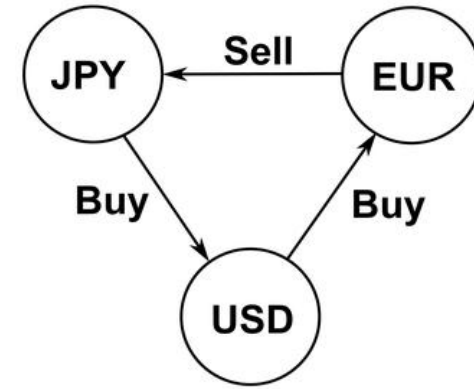
Execution:

Send orders to the exchange,
...

$$EUR/JPY_t < (USD/JPY_t) \times (EUR/USD_t)$$



$$EUR/JPY_t > (USD/JPY_t) \times (EUR/USD_t)$$



Profitable misprices and associated triangular arbitrage strategies

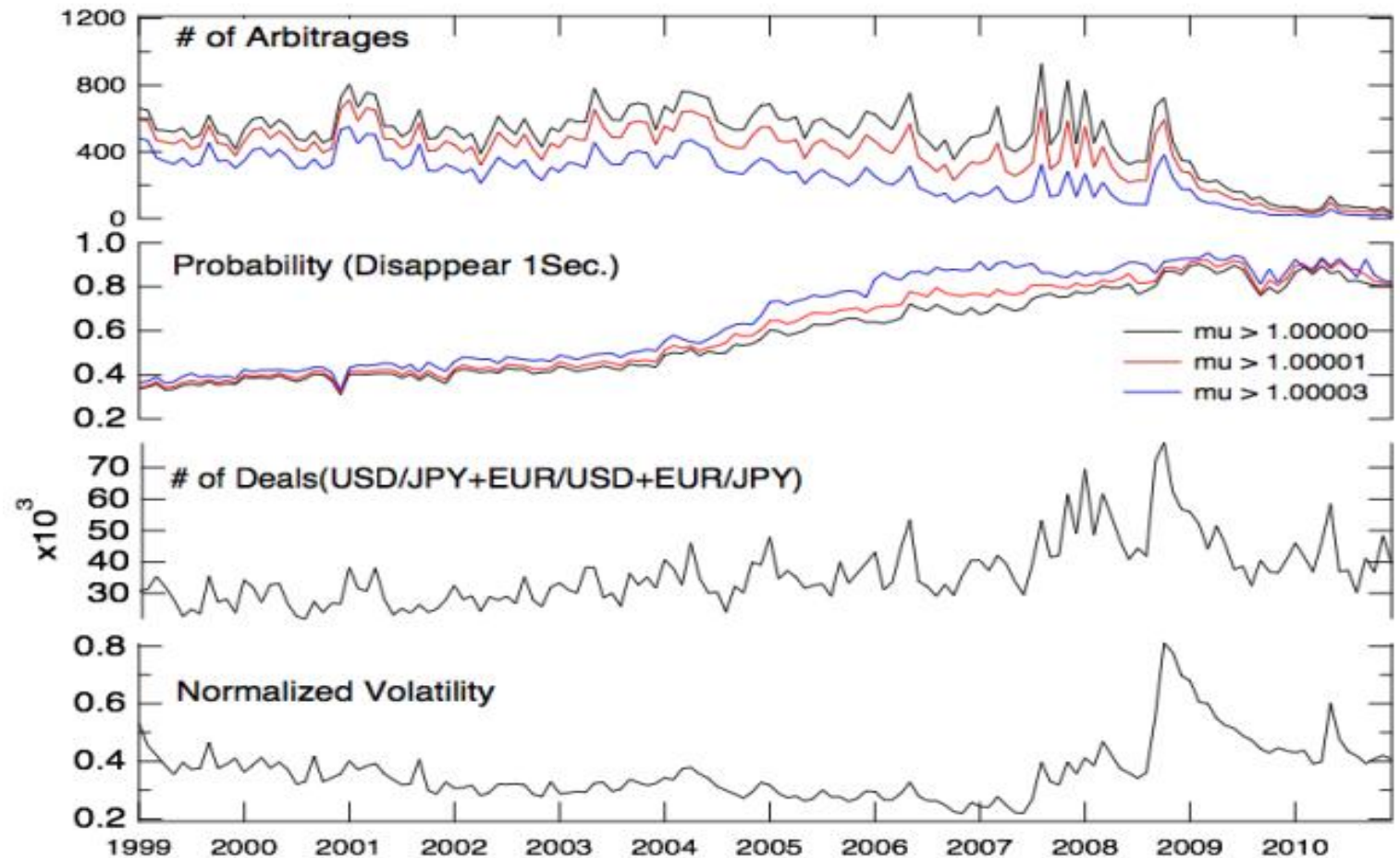
Left : an agent buys EUR for JPY and sells EUR for JPY through USD.

Right : an agent sells EUR for JPY and buys EUR for JPY through USD.

source: Ciacci et al. 2020

2. Co-movements based Quantitative Strategies: Cross-Asset Arbitrage, Triangular Arbitrage

- Speed is crucial!
- Most of trading opportunities (80+%) disappeared within 1 second (13 years ago).
- Crypto market is less competitive but it's getting tougher.



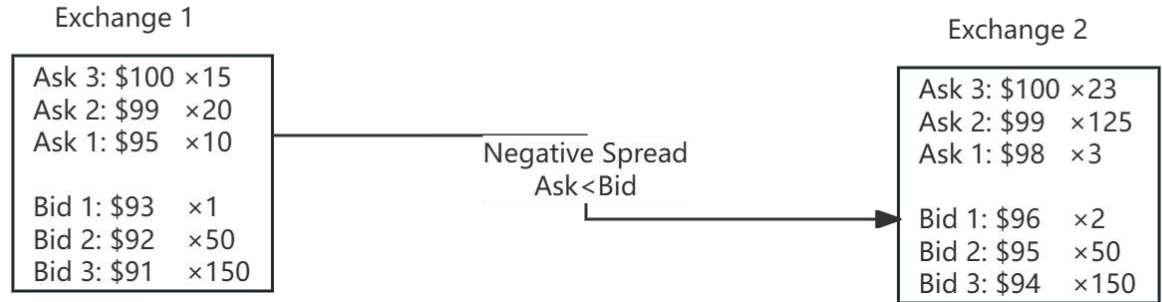
source: Yamada et al. 2012

2. Co-movements based Quantitative Strategies: Cross-Exchange Arbitrage

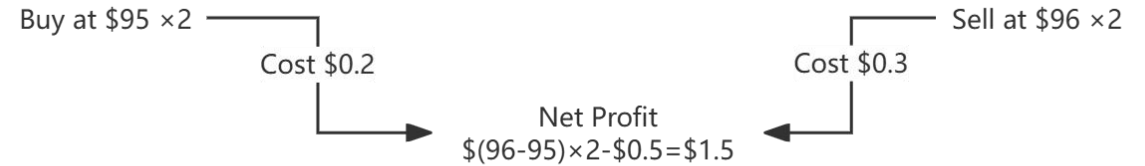
Identify Pairs:

Same Underlying
Same Asset
Different Market

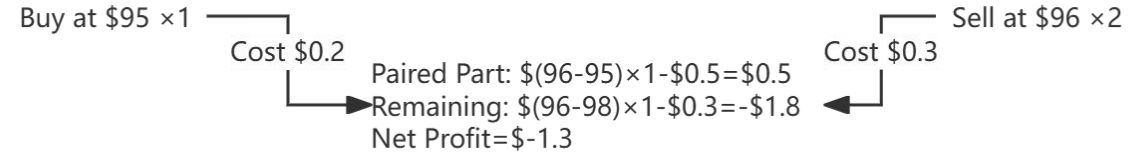
- Speed is crucial! liquidity
- Cryptos have CEX and DEX, which could potentially have more cross-exchange arbitrage opportunities.
- It shall works well on illiquid coins, but has limited capacity.



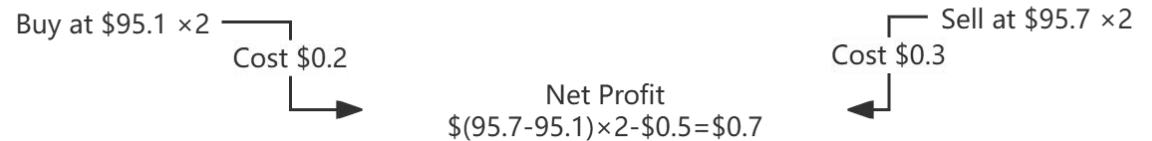
Scenario 1:
Fully Executed



Scenario 2:
Partially Executed



Scenario 3:
Price Priority



3. Modeling Co-movement in Finance: Breadth Indicator

Breadth indicator (BI) measures the overall participation or strength of a market move by assessing how many individual assets are advancing versus declining.

The Core Logic of BI:

A trend with broad participation is robust; one with narrowing participation is weakening and may reverse.

$$BI = \frac{N_{Up_trend}}{N_{Total}}$$

- A generalized BI is flexible in ranges, targeting portfolios, sectors, ETFs, indices, markets, etc.
- Conditions of “Up-trending” are also flexible:
 - Price > MA(T)
 - Price > 52 Week High
 - 50 < RSI(T) < 90



3. Modeling Co-movement in Physics: Complex Networks

- **Complex Network:** The “network” might be thought of as a set of entities (nodes) with a set of pairs from those entities denoted as links which interconnect pairs of nodes, representing some kind of interaction or association between them.
- The main idea behind a complex network is to model high-order interactions among the individual components of large systems by means of graphs, to increase our understanding of not only immediate but also high-order interactions, as well as the implications of these interactions on the mechanisms that govern the underlying phenomenon.
- A market crash is characterized by a sharp rise in connectivity among assets and a corresponding collapse in the average path length of the financial network, creating a small-world topology facilitating for systemic risk spillover.

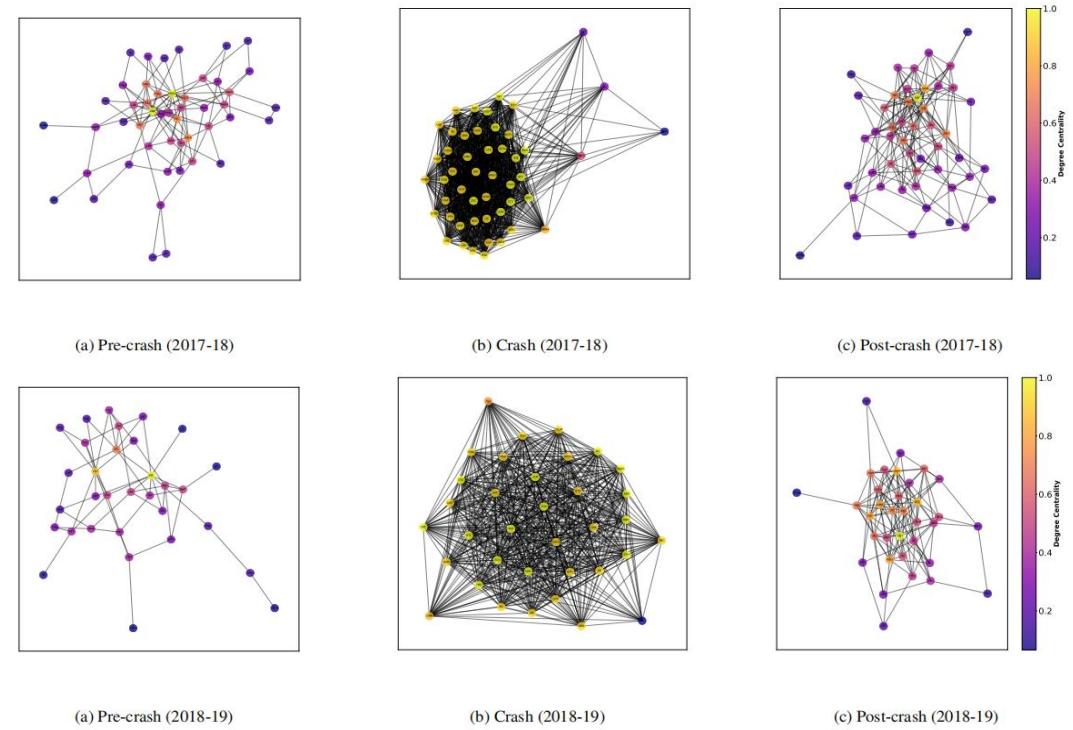


Table 2: Table contains the degree density (ρ_D) of the cryptocurrency network across different crash periods.

Market period	Pre-crash	Crash	Post-crash
2017-18	0.097	0.884	0.181
2018-19	0.115	0.97	0.235
2019-20	0.118	0.25	0.096

- Buyya, R., Calheiros, R. N., & Dastjerdi, A. V. (Eds.). (2016). *Big data: principles and paradigms*. Morgan Kaufmann.
- Mukhia, K., Rai, A., Luwang, S. R., Nurujjaman, M., Majhi, S., & Hens, C. (2024). Complex network analysis of cryptocurrency market during crashes. *Physica A: Statistical Mechanics and its Applications*, 653, 130095.

3. Modeling Co-movement in Physics: Complex Networks

Definition

Node/Vertex $v_i : (x_1, x_2, \dots, x_m)$

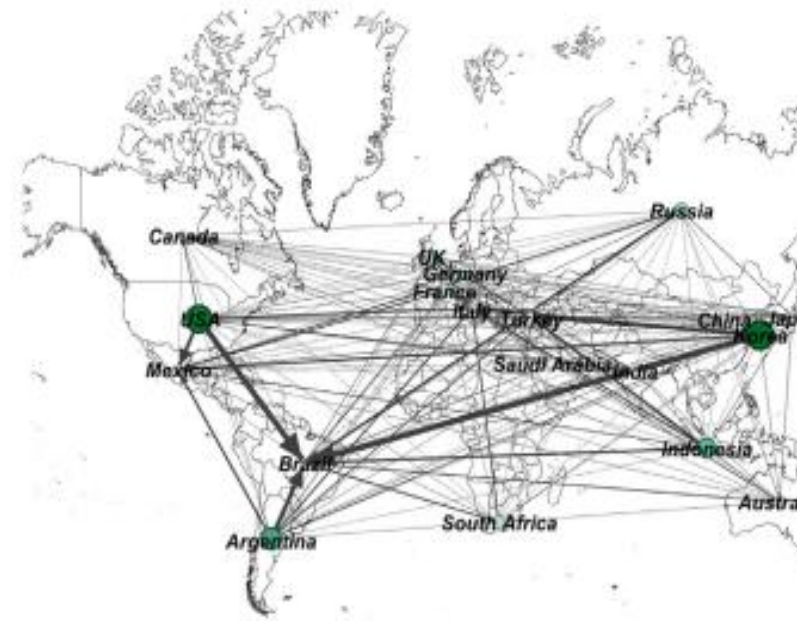
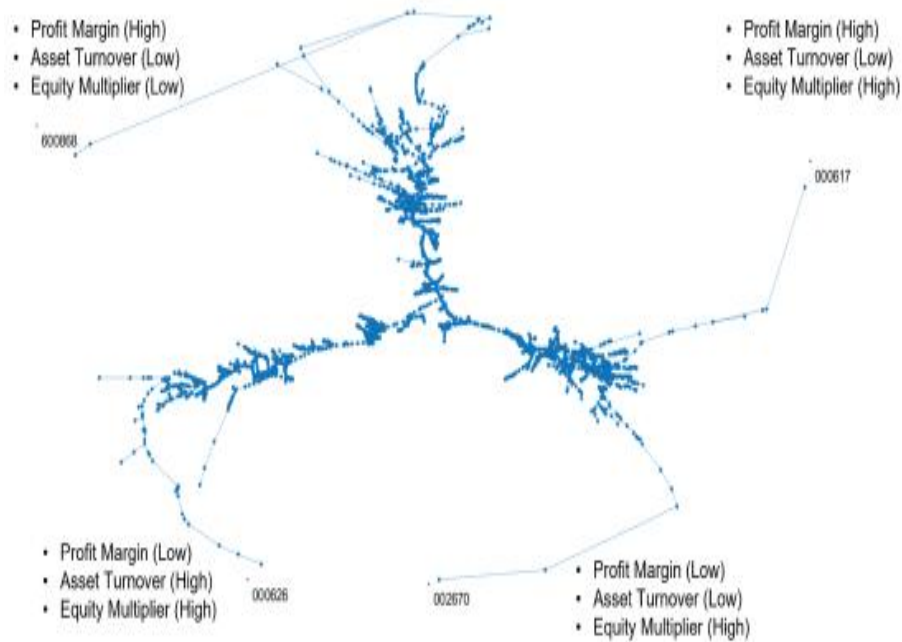
Verticies $V = \{v_1, v_2, \dots, v_n\}$

Adjacency Matrix (Link) $A = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,n} \end{bmatrix}$

Weight Matrix (Edge) $W = \begin{bmatrix} w_{1,1} & w_{1,2} & \dots & w_{1,n} \\ w_{2,1} & w_{2,2} & \dots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n,1} & w_{n,2} & \dots & w_{n,n} \end{bmatrix}$

NETWORK	NODES	LINKS	DIRECTED UNDIRECTED	N	L
Internet	Routers	Internet connections	Undirected	192,244	609,066
WWW	Webpages	Links	Directed	325,729	1,497,134
Power Grid	Power plants, transformers	Cables	Undirected	4,941	6,594
Mobile Phone Calls	Subscribers	Calls	Directed	36,595	91,826
Email	Email addresses	Emails	Directed	57,194	103,731
Science Collaboration	Scientists	Co-authorship	Undirected	23,133	93,439
Actor Network	Actors	Co-acting	Undirected	702,388	29,397,908
Citation Network	Paper	Citations	Directed	449,673	4,689,479
E. Coli Metabolism	Metabolites	Chemical reactions	Directed	1,039	5,802
Protein Interactions	Proteins	Binding interactions	Undirected	2,018	2,930

3. Modeling Co-movement in Physics: Complex Networks



(c) Spillover network in Period 3: 2007–2009.



- Complex networks can serve as an alternative framework to mean–variance analysis for portfolio optimization.
- Complex networks can model and characterize risk spillover and contagion effects in financial markets.
- Yan, X., Yang, H., Yu, Z., & Zhang, S. (2021). A network view of portfolio optimization using fundamental information. *Frontiers in Physics*, 9, 721007.
- Liu, X., An, H., Li, H., Chen, Z., Feng, S., & Wen, S. (2017). Features of spillover networks in international financial markets: evidence from the G20 countries. *Physica A: Statistical Mechanics and its Applications*, 479, 265-278.

Assignment 4: Quant Modeling

Lecture 7,8

- 1.(30) Extend your trading idea into a model. Fill Table 2 (Design of Algorithmic Models) showed in Lecture 5.
- 2.(30) Illustrate why your quant model is a good choice serving the strategy.
- 3.(40) Create clear and specific trading rules based on your quant model outcomes. (When to buy, when to sell, what's the optimized position, etc.)

It is NOT required in Assignment 4 for programming and testing your model. But if you are ready to move forward, try to fill Table 3 and Table 4. It will be Assignment 5. “Forward testing”, “Live Trading”, “Capacity Test” are FYI, you can ignore them during this class.

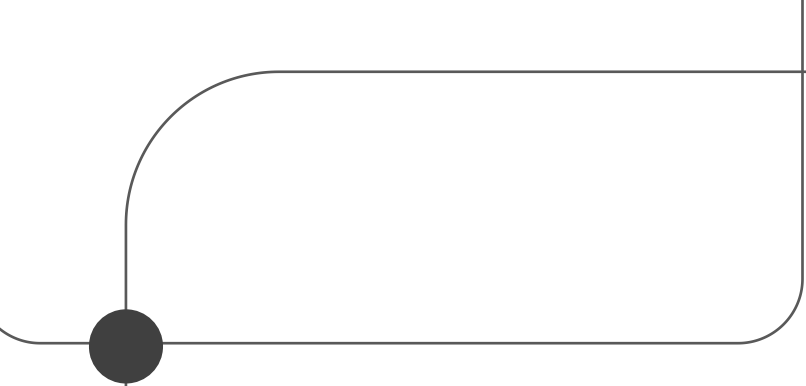
Appendix

Empirical Test (Back Test, Forward Test, Paper Trading, Live Trading)	
Benchmark	
Performance Assessment	Sharpe Ratio, Calmar Ratio, Maximum Drawdown, Expected Return (Annualized, Compounding/Non-compounding), Trading Frequency, Average Holding Period, Winning Probability, etc.
Back Test 1	The back-testing period 1 should contain both bull and bear market conditions.
Back Test 2	The back-testing period 2 should be the nearest 3-6 months to now.
Back Test Scenario 1	Test the strategy performance during typical risk events such as the financial crisis in 2008, trade war, etc.
Back Test Scenario 2	
Paper Trading (Forward Test)	Paper trading is a critical step in strategy development. However, it excludes real-world frictions and operational risks inherent to live markets, such as slippage, market impact, latency, partial order fills, and technical disruptions (e.g., internet disconnections or data feed failures).
Live Trading	
Capacity Test	

Table 3. Empirical Test

Evaluation of Empirical Results	
Refine the Strategy	● Does the strategy’s return, risk, capacity, and trading frequency align with expectations? ● What caused the drawdowns, and are there methods to mitigate them? ● Would earlier or later entry/exit timing improve performance? ● Can risk be reduced or trading frequency increased while maintaining target returns? ● Can the strategy be extended to other instruments or markets to expand opportunities?
Applicable Market Conditions	Observe the best performing market conditions of the strategy. (e.g., high volatility, uptrend with low volume, etc.)
Risk/Position Management	Position control, stoploss/stopgain, hedging, risk redflags (liquidity risk, systemic risk, etc.)
Execution Risk Assessment	Assess the impact of order fill rate, execution latency, slippage, market impact, leverage, and capital gains tax on strategy performance.

Table 4. Evaluation of Empirical Results



Thank You

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